Notes on General Equilibrium Theory

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Chapter 1

Main Concepts, Theorems, and Propositions

1.1 Concepts

1.1.1 Boundary condition

The usual boundary condition used in the existence proof is:

$$p^n \to p \text{ with some } p_l = 0 \Rightarrow \max\{z_1(p^n, w), z_2(p^n, w), dots, z_L(p^n, w)\} \to \infty.$$

1.1.2 Aggregate Excess Demand Function Properties

Let $z(p, w) : \mathbb{R}^L_{++} \times \mathbb{R}^{LI}_{++} \to \mathbb{R}^L_+$ be the excess demand function. It has the following properties ("SWBBH")

1. smoothness: $z(p, w)$ is C^1
2. Walras Law: $pz(p, w) = 0$.
3. bounded from below: $z(p, w) \geq -s$
4. boundary condition: $p^n \to p$ with some $p_l = 0$ then,$\max\{z_k(p, w)\} \to \infty$.
5. homogeneity of degree 0: $z(\lambda p, \lambda w) = \lambda^0 z(p, w) = z(p, w)$.

1.2 Propositions

1.2.1 Sunspots

Intuitively, a sunspot equilibrium means that we can have different equilibrium outcomes even when economic fundamentals are the same.

$$\exists \text{ non-trivial sunspot equilibrium } \Rightarrow \text{ equilibrium indeterminacy or multiplicity.}$$

Cass and Shell. When markets are complete, sunspot equilibria do not exist. That is, if

$$\omega_{s,e} = \omega_{s,e'} = \omega_s,$$

then

$$x_{s,e} = x_{s,e'}.$$

But when markets are incomplete, sunspot equilibria may exist.

Example. Since there is only one bond paying one unit of the numeraire in every state, the asset payoff vector is $A = (1, \dots, 1)'$. Hence, the span condition implies

$$p_s(x_s^i - \omega_s^i) = z^i \qquad \text{for every state } s.$$

After splitting state 1 into two extrinsic states α, β, we have

$$p_\alpha(x_\alpha^i - \omega_1^i) = p_\beta(x_\beta^i - \omega_1^i) = z^i.$$

Thus, the same bond position must finance net trades in both extrinsic states. But this condition only restricts the value of net trades, not the commodity bundle itself. Therefore it does not force $x_\alpha^i = x_\beta^i$.

If equilibrium spot prices differ across the two extrinsic states, $p_\alpha \neq p_\beta$, then the same numeraire payoff can support different consumption bundles. Hence, with incomplete markets, it is possible that

$$x_\alpha \neq x_\beta$$

even though $\omega_\alpha = \omega_\beta = \omega_1$. This is the sunspot logic.

Chapter 2

Math Preliminaries

2.1 Measure Theory

Definition 2.1.1 (Genericity)**.** A property is considered generic (or holds generically) if it applies to a full measure Lebesgue subset of the space of economies. Consider an economy parameterized by $\omega \in \mathbb{R}^{LI}_{++}$. Intuitively, this means that if you were to randomly select an economy (defined by its endowment vector ω), the property would hold with probability 1. Economies that fail to satisfy this property are "rare" (measure zero) and disappear under slight perturbations.

2.2 Differential Topology

Definition 2.2.1 (Transversality)**.** Let $f : \mathbb{R}^M \to \mathbb{R}^N$, with $M > N$. We say f is **transversal** at 0 (denoted as $f \pitchfork 0$), if

$$\forall x, \text{ such that } f(x) = 0$$

, we have the Jacobian matrix

$$Df = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_m} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_m} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_m} \end{bmatrix}$$

is full rank, namely, $\text{rank}(Df) = N$.

Example Consider $M = 2$, and $N = 1$. Define

$$f(x, y) = x^2 + y^2 - 1$$

The zero set is the unit circle. $x^2 + y^2 = 1$.

Since $n = 1$, the Jacobian is a single row vector (the gradient):

$$Df(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} 2x & 2y \end{bmatrix}$$

The rank of a 1×2 matrix is 1 (full rank) unless all entries are zero.

$$Df = [0, 0] \text{ iff } x = 0 \text{ and } y = 0.$$

However, the origin is **not** in the zero set. Therefore, for every point actually on the circle ($f = 0$), the gradient is non-zero. Hence, we have

$$f \pitchfork 0$$

Example 2: Non-transveral

$$g(x, y) = y^2 - x^3$$

The set of points where $y^2 = x^3$. This describes a semicubical parabola which has a sharp "cusp" (singular point) at the origin.

The Jacobian is given by

$$Dg(x, y) = \begin{bmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \end{bmatrix} = \begin{bmatrix} -3x^2 & 2y \end{bmatrix}$$

It follows that $Df(0, 0)$ at the point $(0, 0)$: Therefore, its rank is **0** and thus g is not transversal at 0.

Proposition 1 (Transversality Theorem)**.** *Let $f : \mathbb{R}^M \to \mathbb{R}^N$ be C^∞, with $M > N$. And decompose $x = [x1, x2]$, where $x1 \in \mathbb{R}^{M-N}$, and $x2 \in \mathbb{R}^N$. For almost every $x1$, if $f(x1, x2) = 0$, we have*

$$rank(D_{x2} f) = \begin{bmatrix} \frac{\partial f_1}{\partial x_{2,1}} & \frac{\partial f_1}{\partial x_{2,2}} & \cdots & \frac{\partial f_1}{\partial x_{2,N}} \\ \frac{\partial f_2}{\partial x_{2,1}} & \frac{\partial f_2}{\partial x_{2,2}} & \cdots & \frac{\partial f_2}{\partial x_{2,N}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_N}{\partial x_{2,1}} & \frac{\partial f_N}{\partial x_{2,2}} & \cdots & \frac{\partial f_N}{\partial x_{2,N}} \end{bmatrix} = N.$$

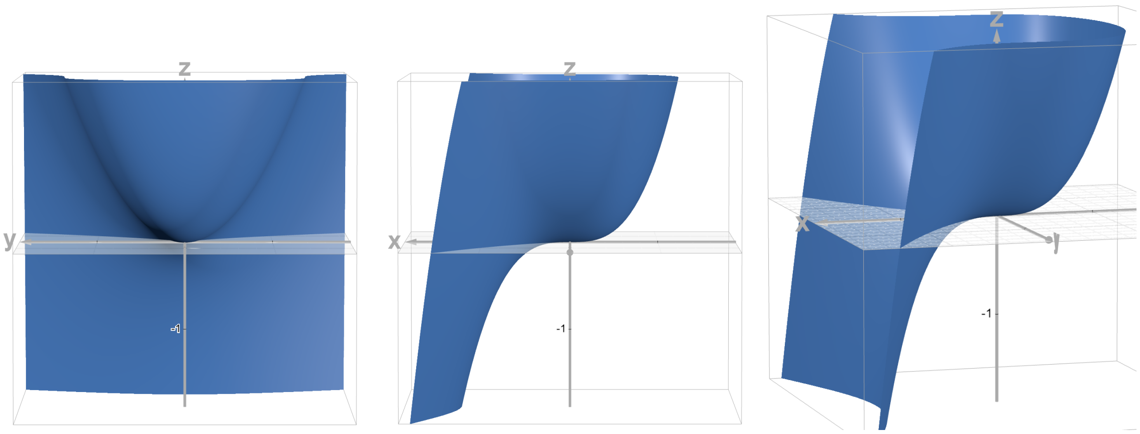

Figure 2.2.1: A function that is not transversal at the origin

NOTES: This function depicts $g(x, y) = y^2 - x^3$, which is not transversal at the origin due to the cusp there.

Remarks: We will use transverality theorem to prove a bunch of important results in GE theory, such as the genericity of regular economies.

Chapter 3

Abstract Exchange Economies

3.1 Pareto Efficient Allocations

Proposition 2 (Negishi Theorem)**.** *(If) Let* $\mathbb{U} = \{U \in \mathbb{R}^I | U \leq U(x)$ *for some* $x \in \mathbb{R}^{LI}$ *such that* $\sum_i x_i \leq \tilde{w}\}$ *be a convex set. Let* x *be a Pareto efficient allocation. Then there exists* $\alpha \in \mathbb{R}^I_+$ *and* $\alpha \neq 0$ *such that* x *is the solution to the following problem:*[1]

$$\begin{aligned} \max_x \quad & \sum_i \alpha_i U_i(x_i) \\ \textit{subject to} \quad & \sum_i x_i \leq \tilde{w} \end{aligned}$$

(Only if) Given $\alpha \in \mathbb{R}^I_+$ *and* $\alpha \neq 0$*, if* x *is the solution to the Negishi Probelm, then* x *is Pareto Optimal.*

Proof (If) Because $\mathbb{U}$ is convex and nonempty, and $U(x) \in \text{brdy}(\mathbb{U})$, by supporting hyperplane theorem, there exists α such that $\alpha U(x) \geq \alpha U'$ for all $U' \in \mathbb{U}$. Hence, $x = \text{argmax}_{x'} \sum_i \alpha_i U_i(x'_i)$. In addition, $\mathbb{U}$ is unbounded from below, we have $\alpha \in \mathbb{R}^I_+$.

(Only if) Suppose $\alpha \in \mathbb{R}^I_{++}$ and x solves the Negishi problem. Suppose, toward a contradiction, that x is not Pareto optimal. Then there exists a feasible allocation x' such that

$$U_i(x'_i) \geq U_i(x_i) \quad \forall i,$$

with strict inequality for some j. Since $\alpha_j > 0$, we have

$$\sum_i \alpha_i U_i(x'_i) > \sum_i \alpha_i U_i(x_i),$$

[1] In other words, there exists a Pareto weights that can justify the allocation.

contradicting the optimality of x in the Negishi problem. Therefore, x is Pareto optimal.
■

3.2 Welfare Analysis

Proposition 3 (First Welfare Theorem). *In an abstract exchange economy characterized by $w \in \mathbb{R}^{S+1}_{++}$, any competitive equilibrium allocation $x \in \mathbb{R}^{S+1}_{+}$ is Pareto efficient.*

Proof: We prove by contradiction. Suppose there exists y that Pareto dominates x, then $y^i \succeq_P x^i \forall i$, and $y^j \succ_P x^j$ for some j. Then by the Walras Law and the Law of Revealed Prferences, we have

$$p_0(y_0^i - w_0^i) + \sum_s p_s(y_s^i - w_s^i) \geq p_0(x_0^i - w_0^i) + \sum_s p_s(x_s^i - w_s^i) = 0, \quad \text{and}$$

$$p_0(y_0^j - w_0^j) + \sum_s p_s(y_s^j - w_s^j) > p_0(x_0^j - w_0^j) + \sum_s p_s(x_s^j - w_s^j) = 0$$

Summing over i (i.e., summing the two equations above), we have

$$p_0 \sum_i (y_0^i - w_0^i) + \sum_s p_s \sum_i (y_s^i - w_s^i) > 0$$

Since p_0, p_s are strictly positive, there exists an s such that

$$\sum_i (y_s^i - w_s^i) > 0,$$

which implies that y is not feasible. ■

The converse of the First welfare theorem is the Second welfare theorem below. **Interpretation:** In words, the Second Welfare Theorem states that any Pareto efficient allocation can be decentralized as a competitive equilibrium via transfer (equivalently, redistribution of initial endowment, where the transferred endowment $\Delta w^i = \frac{T_i}{p}$) — See MWG Figure 15.b.13 for an illustration.

Proposition 4 (Second Welfare Theorem). *For any economy characterized by $w \in \mathbb{R}^{LI}_{++}$, any Pareto efficient allocation $x \in \mathbb{R}^{LI}_{+}$ can be decentralized as a competitive equilibrium via price, $p \in \mathbb{R}^{L}_{++}$, under redistributed endowments with $\sum_i w_i + t_i = w$ (i.e., $\sum_i t_i = 0$).*

Proof Remark: The key of the proof is a separating hyperplane argument.
Let $x \in \mathbb{R}^{LI}_{++}$ be a Pareto Efficient allocation. Choose $w = x$. Define the strict upper contour

set for agent i,

$$B^i(x_i) := \{y_i \in \mathbb{R}^L_+ : U(y_i) > U(x_i)\}$$

Define the aggregate strict upper contour set as

$$B(x) := \left\{ \sum_i y_i \in \mathbb{R}^L_+ : y_i \in B^i(x_i) \forall i \right\}$$

Since $B^i(x_i)$ is convex, the aggregate upper contour set is also convex. In addition, by Supporting Hyperplane theorem, there exists a nonzero price vector p, such that

$$\inf_{a \in B(x)} pa \geq p \sum_i x_i$$

We will use this equation to establish that

(1) $p \in \mathbb{R}^L_{++}$, and

(2) under such price, no individual will trade.

Part 1. Suppose there exists some l such that $p_l < 0$, let

$$y^i_k = \begin{cases} x^i_k + \delta & \text{if } k = l \\ x^i_k & \text{if } k \neq l \end{cases}$$

Then, $py^i < px^i$ and $y^i \in B^i(x^i)$. Summing across agents, we have a contradtion to the supporting hyperplane condtion above.

Now suppose there exists some l such that $p_l = 0$, let

$$y^i_k = \begin{cases} x^i_k + \delta & \text{if } k = l \\ x^i_k - \epsilon & \text{if } k \neq l \end{cases}$$

where δ is sufficiently large and ϵ is sufficiently small, so that the loss in $k \neq l$ can be compensated. Then, $py^i < px^i$ and $y^i \in B^i(x^i)$. Summing across agents, again, we have a contradiction to the supporting hyperplane condtion above.

Part 2. Finally, we prove that under p found by the supporting hyperplane theorem, the equilibrium is autarky. Suppose there is a strictly profitable deviation from x^i for agent i, namely,

$$\exists \tilde{x}^i, \quad \text{s.t. } p\tilde{x}^i < px^i \quad \text{and} \quad \tilde{x}^i \in B^i(x^i)$$

Denote the payoff from the deviation by $\delta := px^i - p\tilde{x}^i > 0$. Pick $\tilde{x}^i + \sum_{j \neq i} y^j \in B(x)$, and apply the SHT, to obtain

$$p\left(\tilde{x}^i + \sum_{j\neq i} y^j\right) \geq p\sum_i x^i$$

It follows that

$$p\left(\sum_{j\neq i} y^j - x^j\right) \geq \delta > 0$$

However, consider $y^j = x^j + \epsilon$, we have $\lim py^j = \lim px^j$ as $\epsilon \to 0$, a contradiction. Hence, it cannot exist any profitable deviation for any agent i. ■

3.3 Sufficient conditions for the uniqueness of competitive equilibrium

3.4 Sonnenschein-Debreu-Mantel Theorem (Anything-goes)

The following famous theorem is known as the "anything-goes" theorem and Indeterminacy Theorem, in the sense that, one can take "any" arbitrary aggregate excess demand function (z), and construct an economy with rational individuals that matches it. This proves that individual rationality imposes no restrictions on the aggregate market behavior. This undertermineness at the macro level outcomes leads to a famous quote by Tom Sargent:

> "The real problem in economics is not that our models impose too many restrictions on the data, but far too few".

Jesús Fernández-Villaverde says: SMD teaches us that meaningful economic predictions require disciplined structure beyond individual rationality.

Proposition 5 (Sonnenschein-Debreu-Mantel Theorem). *Let Δ_ϵ^{L-1} be a trimmed simplex, with $\epsilon > 0$. Given any continuous function $z : \Delta_\epsilon^{L-1}$ satisfying continuity and Walras law, then there is an economy with $U^i : \mathbb{R}_+^L \to \mathbb{R}$, $\omega \in \mathbb{R}_{++}^{LI}$, $I \geq L$, whose aggregate excess demand function, restricted to Δ_ϵ^{L-1}, equals z.*

3.5 Consumption Externalities and Pigouvian Taxes

In the presence of consumption externalities, an individual's utility depends on both her and others' consumptions. As a result, the centralized allocation (i.e., Negishi's solution) deviates from the decentralized one. To induce the decentralized allocation to be Pareto Efficient, one can introduce the so-called Pigouvian taxes, so that the taxes make individuals behave *as if* they care about others. The following derives the tax formula.

For simplicity, we assume that the goods 1 does not have consumption exteranlity. First, the Negishi problem is give by[2]

$$\begin{aligned} \max_{x \in \mathbb{R}^{LI}} \quad & \sum_i \alpha^i U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) \\ \text{subject to:} \quad & \sum_i x_l^i \leq w_l, \quad \forall l \end{aligned}$$

Note that there are L constraints, so we introduce L Lagrangean Multipliers in the equation below

$$\mathcal{L} = \sum_i \alpha^i U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) + \sum_l \lambda_l \left(w_l - \sum_i x_l^i \right)$$

The first-order conditions are given by

$$\begin{aligned} [x_1^i] \qquad & \alpha^i \frac{\partial U^i}{\partial x_1^i} = \lambda_1 \\ [x_l^i] \qquad & \sum_j \alpha^j \frac{\partial U^j}{\partial x_l^i} = \lambda_l \quad \text{ for } l \neq 1 \end{aligned}$$

The decentralized consumer's problem is given by

$$\begin{aligned} \max_{x^i \in \mathbb{R}^I} \quad & U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) \\ \text{subject to:} \quad & \sum_l p_l(1+\tau_l^i)x^i \leq \sum_l p_l w_l^i, \quad \forall l \end{aligned}$$

The Lagrangean is given by

$$\mathcal{L}^i = U^i(x_1^i, \{x_l^i\}, \{x_l^{-i}\}) + \mu^i \sum_l \left[p_l w_l^i - p_l(1+\tau_l^i)x^i \right]$$

[2] We use superscripts to denote individuals, and subscripts to index goods.

The first-order conditions are given by

$$
\begin{aligned}
&[x_1^i] \qquad \frac{\partial U^i}{\partial x_1^i} = \mu^i p_1 = \mu^i \qquad \text{normalization using} \quad p_1 = 1 \\
&[x_l^i] \qquad \frac{\partial U^i}{\partial x_l^i} = \mu^i p_l(1+\tau_l^i) \quad \text{for } l \neq 1
\end{aligned}
$$

Combining the these two conditions, we have

$$\frac{\partial U^i}{\partial x_l^i} = \frac{\partial U^i}{\partial x_1^i} p_l(1+\tau_l^i)$$

Note that we can decompose the second FOC of the Negishi problem as

$$\frac{\partial U^i}{\partial x_l^i} + \sum_{j\neq i} \frac{\alpha^j}{\alpha^i}\frac{\partial U^j}{\partial x_l^i} = \frac{\lambda_l}{\alpha^i} \qquad \text{for } l \neq 1$$

Therefore, we can equate the equation above with the decentralized condition and compute the tax formula:

$$\underbrace{\frac{\lambda_l}{\alpha_i} - \sum_{j\neq i} \frac{\alpha^j}{\alpha^i}\frac{\partial U^j}{\partial x_l^i}}_{\text{Negishi's}} = \underbrace{\frac{\partial U^i}{\partial x_1^i} p_l(1+\tau_l^i)}_{\text{Decentralized consumer's}}$$

It follows that

$$\tau_l^i = \frac{\lambda_l}{p_l \alpha_i \frac{\partial U^i}{\partial x_1^i}} - 1 - \sum_{j\neq i} \frac{\alpha^j}{p_l \alpha^i} \frac{\frac{\partial U^j}{\partial x_l^i}}{\frac{\partial U^i}{\partial x_1^i}}$$

Now assuming $\alpha^i = \frac{1}{\mu^i} = \frac{1}{\frac{\partial U^i}{\partial x_1^i}}$, and $p_l = \lambda_l$, the we have

$$\boxed{\tau_l^i = -\frac{1}{p_l}\sum_{j\neq i} \frac{\frac{\partial U^j}{\partial x_l^i}}{\frac{\partial U^j}{\partial x_1^j}}.}$$

Chapter 4

Financial Market Economies

4.1 Arrow-Debreu Equilibrium

Definition 4.1.1. An Arrow-Debreu equilibrium is defined by an allocation $x_0, \{x_s\}$, such that

1. given the goods prices ϕ_0, ϕ_s, the allocation solves the consumer's problem below:

$$\begin{aligned} \max_{x_o, \{x_s\}} \quad & u_0(x_0) + \sum_s \text{Prob}_s u_s(x_s) \\ \text{s.t.} \quad & \phi_0(x_0 - w_0^i) + \sum_s \phi_s(x_s - w_s^i) = 0 \end{aligned}$$

2. Goods market clears:

$$\sum_i x^i \leq \sum_i w^i$$

4.2 No-Arbitrage

There are multiple definitions of the No Arbitrage condition. Essentially, they all require zero profit on the spot market at time 0, and the future financial market no matter what the future state will be. Formally, the first definition is given by

Definition I (No Arbitrage): Let $A \in \mathbb{R}^{S\times J}$ be a future return matrix. Let $z \in \mathbb{R}^J$ be a portfolio. $q^T \in \mathbb{R}^J$ be a price vector of the portfolio.[a] For $W = \begin{bmatrix} -q \\ A \end{bmatrix}$, there is no $z \in \mathbb{R}^J$ such that $Wz > 0$.

- **Remark:** Intuitively, $Wz \in \mathbb{R}^{S+1}$ denotes the profit of buying/selling assets on the spot market and all the future financial markets.

[a] In this note, only q is a row vector, while other vectors are columns.

Example: Consider $W = \begin{bmatrix} -3 & -4 \\ 5 & 0 \\ 2 & 2 \\ 3 & 1 \end{bmatrix}$. Then take $z = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ (namely, buying 1 unit of asset 1 and selling 1 unit of asset 2), we have $Wz = \begin{bmatrix} 1 \\ 5 \\ 0 \\ 2 \end{bmatrix}$, which means the agent can make profit 1 at time 0 (on the spot market), and 5, 0, 2 at time 1 if $s = 1, 2, 3$, respectively.

The No Arbitrage condition above is (mathematically) equivalent to the following definition

Definition II (No Arbitrage):

$$\text{span}(W) \cap \mathbb{R}_+^{S+1} = \{\mathbf{0}\},$$

where $\text{span}(W) = \{Wz, \forall z \in \mathbb{R}^J\}$ denotes the profit of all posible portfolio.

With these two definitions in mind, we now introduce the No-Arbitrage theorem

Defintion (No-Arbitrage Theorem)

$$\text{span}(W) \cap \mathbb{R}_+^{S+1} = \{\mathbf{0}\} \Leftrightarrow \exists \hat{\pi} \in \mathbb{R}_{++}^{S+1} \text{ such that } \hat{\pi}\tau = 0, \forall \tau \in \text{span}(W)$$

Remarks:

- We require $\hat{\pi} \in \mathbb{R}_{++}^{S+1}$ (strictly positive) due to the need of normalization (see below).

Intuitively, $\hat{\pi}$ serves as a machinery that summarizes the agent's profit at time 0 and time

1, taking probabilities and discount factors into considerations:

$$\hat{\pi}_0[Wz]_0 + \sum_s \hat{\pi}_s[Wz]_s = 0, \quad \forall z \in \mathbb{R}^J$$

Since $[Wz]_0 = -\sum_j q_j z_j$, after the normalization $\pi_s = \frac{\hat{\pi}_s}{\hat{\pi}_0}$, we have

$$-\sum_j q_j z_j + \sum_s \pi_s \sum_j a_{sj} z_j = 0,$$

where π_s denotes the price of one unit of consumption good delivered in state s at time 1. In matrix form, we have $(-q + \pi^T A)z = 0 \Rightarrow q = \pi^T A$.

Defining $\pi_s = m_s \text{Prob}_s$, we have the **Fundamental Equation of Asset Pricing**:

$$q = \mathbb{E}\left(mA\right). \tag{4.2.1}$$

Defintion (No-Arbitrage Theorem)

$$\text{span}(W) \cap \mathbb{R}^{S+1}_{+} = \{\mathbf{0}\} \Leftrightarrow \exists \hat{\pi} \in \mathbb{R}^{S+1}_{++} \text{ such that } \hat{\pi}\tau = 0, \forall \tau \in \text{span}(W)$$

Proof of the No-Arbitrage Theorem

($\Rightarrow$) We prove this direction in two steps:

1. "Strict positivity". First, we define a simplex in the space of $\mathbb{R}^{S+1}$:

$$\Delta = \{\tau \in \mathbb{R}^{S+1} : \sum_{s=0}^{S} \tau_s = 1\}$$

By the definition of No-Arbitrage condition (i.e., the LHS of the theorem), we have

$$\text{span}(W) \cap \Delta = \emptyset$$

Now, invoking the **Strong Separating Hyperplane Theorem** (this is the essence of the proof), then there exists $\hat{\pi}$, such that

$$\sup_{\tau \in \text{span}(W)} \hat{\pi}\tau < \inf_{\tau \in \Delta} \hat{\pi}\tau \tag{4.2.2}$$

To prove $\hat{\pi} \in \mathbb{R}^{S+1}_{++}$, suppose there exists a $\hat{\pi}$ whose s^{th} component, $\pi_s \leq 0$, then since the base vector ($e_s \in \Delta$), we have

$$\inf_{\tau \in \Delta} \hat{\pi}\tau \leq e_s\tau \leq 0$$

However, $0 \in \text{span}(W)$ (in words, no buying and selling in the asset market), which implies that

$$\sup_{\tau \in \text{span}(W)} \hat{\pi}\tau \geq \hat{\pi}0 = 0$$

Hence, we have the $0 \leq LHS < RHS \leq 0$ in equation (**??**), a contradiction.

2. "Orthorgonality".. Suppose there exists a $\tau \in \text{span}(W)$ such that $\hat{\pi}\tau \neq 0$. Then, we can find an α such that $\hat{\pi}\alpha\tau$ is arbitrarily large. (In words, agent can make infinite profit if there is an opportunity for arbitrage). However, the RHS of the theorem is bounded above (because Δ is a compact set). Then we have a contradiction. Hence,

$$\hat{\pi}\tau = 0 \forall \tau \in \text{span}(W)$$

($\Leftarrow$) Now suppose there exists an arbitrage opportunity $\tau^* \in \text{span}(W)$ such that $\tau^* \in \mathbb{R}^{S+1}\backslash\{0\}$. Since $\hat{\pi} \in \mathbb{R}^{S+1}_{++}$, we have $\hat{\pi}\tau^* > 0$. However, this contradicts to the assumption that $\hat{\pi}\text{span}(W) = 0$ as $\tau^* \in \text{span}(W)$. ■

Proposition. (The degree of freedom of stochastic discounts) Given $A \in \mathbb{R}^{S\times J}$ of rank J and $q \in \mathbb{R}^J$, the set of viable stochastic discounts is an $\mathbb{R}^{S-J}$ strictly positive subspace of $\mathbb{R}^J$, namely,

$$R(q; A) = \{\pi \in \mathbb{R}^S_{++} : q = \pi^T A\} = \mathbb{R}^{S-J}_{++}$$

In words, once you find a viable π, there are only $S - J$ elements in that π you can vary.

4.3 Arrow Theorem

4.3.1 Budget constraints equivalence

Proposition 6. *The following two budget constraints are equivalent under No-Arbitrage:*

1. $p_0(x_0^i - w_0^i) + qz = 0$ *and* $p_s(x_s^i - w_s^i) = a_s z$
2. $p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s(x_s^i - w_s^i) = 0$ *and,*

$$\begin{bmatrix} \vdots \\ p_s(x_s^i - w_s^i) \\ \vdots \end{bmatrix} \in span(A)$$

Remarks: When the asset market is complete $\text{span}(A) = \mathbb{R}^J = \mathbb{R}^S$.

Proof Suppose q satisfies the No-Arbitrage condition, we have the following asset pricing equation

$$q = \pi A, \quad \text{with } q \in \mathbb{R}^J, \pi \in \mathbb{R}^S, A \in \mathbb{R}^{S\times J}$$

Now the first equation of the first set of constraints becomes

$$p_0(x_0^i - w_0^i) + qz = p_0(x_0^i - w_0^i) + \pi Az = 0,$$

which is equivalent to

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s a_s z = 0$$

Using the the second equation of the first set of constraints to replace $a_s z$, we have

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s(x_s^i - w_s^i) = 0$$

To complete the proof, notice that the second equation of the second set of constraints is simply the matrix for of

$$p_s(x_s^i - w_s^i) = a_s z$$

$\Rightarrow$

$$\begin{bmatrix} \vdots \\ p_s(x_s^i - w_s^i) \\ \vdots \end{bmatrix} = \begin{bmatrix} \vdots \\ a_s \\ \vdots \end{bmatrix} z$$

which belongs to the span of A. ■

When $L = 1$, we can use the above statement about "constraint equivalence" to show that the constrained set of consumption $B^i \perp p$, and thus we have constrained P.E. Namely,

$$\left(B^i := \begin{cases} x_0^i - w_0^i + \sum_{s=1}^S \pi_s(x_s^i - w_s^i) = 0 \\ [(x_s^i - w_s^i)]_{\forall s} \in \langle A \rangle \end{cases} \right) \perp q$$

Notice that we use $q = \pi A$ in the derivation of the first equation of B^i. However, in the presence of transaction costs, $q \leq \pi A \leq q + \gamma$, the tricks we used in the proof of Proposition **??** break down. Hence, $B^i(q)$ is a function of q when $\gamma > 0$.

4.3.2 Arrow's Theorem

Theorem 1 (Arrow's Theorem). *Under the No-Arbitrage condition, any Arrow-Debreu equilibrium* $(x_0^*, \{x_s^*\}, \{p_0^*, p_s^*\})$ *can be supported as a Financial Market Equilibrium (FME)* $(x_0^*, \{x_s^*\}, z^*, \{p_0^*, q^*\})$*, where* z^* *is the portfolio that satisfies the budget constraints in the FME, and* q^* *satisfies the No-Arbitrage condition.*

Proof Since goods market clearing implies financial market clearing, the market clearing conditions in the FME is the same as the one in Arrow-Debreu equilibrium. Now, we only need to discuss the equivalence of the budget constraints between these two economies.

Since the market in complete, the span condition in the FME is not binding. Then the budget constraints in the FME reduce to

$$p_0(x_0^i - w_0^i) + \sum_s \pi_s p_s(x_s^i - w_s^i) = 0$$

Now using $p_0 = \phi_0$ and $p_s = \pi_s \phi_s$, we can immediately see the equivalence between the AD equilibrium and the FME. ■

4.4 Non-convexity

Proposition 7. *In economies with non-convex commodity space, an equilibrium may fail to exist.*

Proof (Sketch): Consider an economy with non-covex commodity space, for example, when the quantities of some goods are discrete (namely, indivisible goods). Then, the aggregate excess demand function may not be continuous. To tackle this issue, one can consider the convex hull of the aggregate excess demand function. By Kakutani's FPT, we have $0 \in \text{conv}(Z(\pi, p))$. If $0 \in Z(\pi, p)$, then we are done. Otherwise, we can apply the Caratheodory's theorem to find $S + 1 + 1$ vector such that $\sum \alpha_k z_k = 0$ with $z_k \in Z(\pi, p), \alpha_k \geq 0, \sum \alpha_k = 1$. Since agent i is indifferent among $\{z_{ik}\}$ across k, we can assign α_k fraction of type i agents to choose z_{ik}. As a result, the aggregate excess demand is zero. ■

4.5 Cass Trick

Question: What is the Cass trick and how is it useful in the existence proof for (single commodity, two-periods) financial market economies with incomplete markets?

Answer: The definition of the Cass trick. Consider an economy where the span condition is imposed on all consumers except $i = 1$. Then, the Cass trick aruges that the Financial market equilibrium (FME) is the same as the "modified" FME. We use the conclusion of the Cass trick to establish the boundary condition of the FME. Specifically, for $i = 1$, her excess demand goes to infinity as price goes to zero. Using the budget constraints for other individuals, we can see that the the excess demand is bounded from below. Hence, using the same proof as in the existence proof of the AD-equilibrium, we can establish the existence.

4.6 Cass Trick in a 3-period FME

Want: The existence proof of a competitive equilibrium in a 3-period Financial Market Equilibrium.

Sketch: First, we can extend the Lemma of constraint equilvance to a 3-period setup, which includes $1 + S + S^2$ budget constraints (i.e., normalizations). The S^2 additional equations come from the **spanning condition** at period $t = 1$. This extended theorem reduces the price space to $(\pi, p) \in \mathbb{R}_{++}^{S+S^2} \times \mathbb{R}_{++}^{L(1+S+S^2)}$, where π is the state price and p is the spot price.

In addition, the market clearing conditions are

$$\sum_i x^i(\pi, p) - w^i = 0$$

The total number of market clearing conditions are $L(1 + S + S^2) - 1$ because there are L goods, 1 spot market today, S tomorrow and S^2 the day after tomorrow. -1 because of the Walras Law.

Then we can count the total number of unknowns and equations:

$$\underbrace{S + S^2 + L(1 + S + S^2)}_{\text{raw unknowns}} - \underbrace{1 + S + S^2}_{\text{normalization}} = \underbrace{L(1 + S + S^2) - 1}_{\text{\# of mkt clearing cond.}}$$

The role played by the Cass trick is to eliminate the spanning conditions in $t = 1, 2$ for agent $i = 1$. So that the boundary condition holds and the equilibrium exists under this modified equilibrium. The last piece of the proof is just to show that the equilvalence between the modified and the original equilibriua.

4.7 Aggregation

As many economic analysis are conducted under the representative agent framework, such as the canonical consumption-based CAPM model, it is important to understand under what conditions we can aggregate agents' behavior.

Proposition 8. *If utility functions are identitical and homethetic, we have the equilibrium prices* ***independent*** *of the distribution of endowments across agents. (In other words, p is a functionm of only aggregate endowmenet $\sum_i w^i$.)*

Proof of proposition ?? (Sketch): Define demand function $x_s^i(p_s, y_s^i)$. By Walras' law, we have $p_s x_s^i(p_s, y_s^i) = y_s^i$, and $\sum_i y_s^i = p_s \sum_i x_s^i = p_s \sum_i w_s^i$. Then, we have

$$
\begin{aligned}
& \sum_i p_s x_s^i(p_s, y_s^i) = \sum_i y_s^i \\
\Rightarrow \quad & \sum_i p_s x_s(p_s, y_s^i) = p_s \sum_i w_s^i \\
\Rightarrow \quad & p_s \sum_i y_s^i x_s(p_s, 1) = p_s \sum_i w_s^i \\
\Rightarrow \quad & \sum_i y_s^i x_s(p_s, 1) = \sum_i w_s^i \\
\Rightarrow \quad & x_s(p_s, \sum_i y_s^i) = \sum_i w_s^i \\
\Rightarrow \quad & x_s(p_s, p_s \sum_i w_s^i) = \sum_i w_s^i
\end{aligned}
$$

Therefore, p_s is uniquely determined by the equation above. ■

Chapter 5

General Equilibirum with Incomplete Markets (Primer)

5.1 Constrained PE in production economies with incomplete markets ($L = 1$)

5.1.1 Takeaway

Consider a financial market economy that is already Constrained Pareto efficient, adding production and equity trades

- does NOT induce inefficiency if firms adopt Minkowski's rational price conjecture;
- induces inefficiency under Drérez price conjecture.

5.2 Pecuniary externalities and constrained inefficiency

In this section, we introduce several examples of the so-called "Pecuniary Externalities", where the prices have dual roles in the economy:

- They clear the markets;
- They also affect the restrictions on agents' choice set. Concretely, the consumption allocation is restricted by:

$$\begin{bmatrix} \vdots \\ x_s^i \\ \vdots \end{bmatrix} \in B^i(p_1, \ldots, p_S).$$

An important proposition that relates to the second role of prices in the GEI is the following:

Proposition 9. *Constrained Pareto Efficiency* $\Rightarrow B^i \perp p$.

Equivalently, If B^i depends on prices p, the equilibrium allocation is generally Constrained Inefficient.[a]

[a] An exception is homothetic preferences.

The converse is not true for the proposition above if $L > 1$. But by the Diamond's theorem, we have if $L = 1$ (single good economy), then

Proposition 10. $B^i \perp p \Leftrightarrow$ *Constrained Pareto Efficiency* *when* $L = 1$.

Consider an economy with agents $i \in \mathcal{I}$, two dates $t = 0, 1$, and states $s \in \mathcal{S}$ at date 1 with probabilities π_s. Let $z \in R^K$ be the asset portfolio vector with date-0 price vector q. The asset payoff in state s is $a_s \in R^K$.

Consumer i's Problem:

$$\begin{aligned} \max_{\{x^i, z^i\}} \quad & u^i(x_0^i) + \sum_{s \in \mathcal{S}} \pi_s u^i(x_s^i) \\ \text{s.t.} \quad & p_0(x_0^i - w_0^i) + q \cdot z^i = 0 \\ & p_s(x_s^i - w_s^i) = a_s \cdot z^i, \quad \forall s \in \mathcal{S} \end{aligned}$$

5.3 Default

5.3.1 Environment

Consider an economy with I agents, S states, a single commodity ($L = 1$), and a full set of Arrow securities: $A = I_S$ (where I_S is an S-dimensional identity matrix.) Agents can default in each state $s \in$, after the state is realized. If they default they consume only a fraction $\alpha \in (0, 1)$ of their endowment. The remaining fraction, $1 - \alpha$, is first pooled across all defaulting agents and then redistributed pro-rata to their creditors.

5.3.2 Equilibirum

Definition: A competitive equilibrium with default under complete financial market is defined by asset prices ($q \in \mathbb{R}^S_{++}$), consumption choices $x^i \in \mathbb{R}_{++}$, and default choices

$d^i \in \{0, 1\}$, such that

1. Each consumer solves

$$\max_{x_0^i, x_s^i, d^i} \quad u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i)$$

subject to: $\quad x_0^i - w_0^i + qz^i = 0$

$$\forall s \in S, \quad x_s^i = \begin{cases} w_s^i + \underbrace{\frac{z_s^i}{\sum_i \max\{0, z_s^i\}}}_{\text{pro-rata weight}} \sum_{j \neq i} \left[d_j (1-\alpha) w_s^j + (1-d_j) \underbrace{\max\{0, -z_s^j\}}_{\text{identifying borrowers}} \right], & \text{if } z_s^i \geq 0 \\ \\ d_i \alpha w_s^i + (1-d_i)\left(z_s^i + w_s^i\right), & \text{if } z_s^i < 0 \quad (i \text{ is a borrower}) \end{cases}$$

where I use the state index s to denote the asset as the financial market is complete.

2. Goods market clears

$$\sum_i x^i = \sum_i w^i$$

3. Asset market clears

$$\sum_i z^i = 0$$

5.3.3 Efficiency

Consider a situation where $J = S - 1$, $L = 1$, and no-default constraints are imposed on assets $j = 2, 3, \ldots, S-1$.

- Is competitive equilibrium constrained Pareto Efficient?

Note that the no-default condition is

$$w_s^i + z_s^i > \alpha w_s^i$$

$\Rightarrow$

$$z_s^i > -(1-\alpha) w_s^i$$

Now consider the budget constraint:

$$x_0^i - w_0^i + \sum_s q_s z_s^i = 0$$

$$x_s^i - w_s^i = z_s^i$$

Using No-Arbitrage condition, we have $q = \pi A$, As a result, the budget constraint becomes

$$x_0^i - w_0^i + \sum_s \pi_s (x_s^i - w_s^i) = 0 \quad \text{and} \quad \underbrace{\begin{bmatrix} \vdots \\ x_s^i - w_s^i \\ \vdots \end{bmatrix} \in \mathbb{R}^{S-1} \cap \text{ (No-Default constrained set)}}_{\text{Doesn't depend on prices}}$$

Since $L = 1$, the constained set doesn't depend on prices of goods or assets, the competitive equilibrium is **constrained Pareto Efficient**.

Chapter 6

Sequential Production Economies

Timeline

Firms buy at time 0 on the spot market and sell at time 1 on the state-contingent markets.

6.1 Takeaways

- Arrow-Debreu equilibrium
- Financial Market equilibrium
- Minkowski's rational price conjecture and Dreze conjecture
- The most important proof: (L = 1, H = 1, J = 0)
  <u>Competitive equilibria of FME with production under Rational Conjectures are Constrained P.E.</u>
- Corporate finance (e.g., equity vs. bonds)

Notations

- δ^{ih}: endowment of firms' shares
- ϕ: prices
- θ: quantity of equity trade

6.2 Arrow-Debreu Equilibrium (with Production)

Definition.
An Arrow-Debreu Equilibrium is $(x, y, \phi) \in \mathbb{R}_{+}^{L(S+1)I} \times \Upsilon \times \mathbb{R}_{++}^{L(S+1)}$ (N.B.: $y = [y_h]$, with $y_h = (-k^h, f^h(k^h))$.) such that

1. Given ϕ, each individual solves the utility maximization pb:

$$\begin{aligned} \max_{x_0^i, x_s^i} \quad & u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i) \\ \text{subject to:} \quad & \phi_0(x_0^i - w_0^i) + \sum_s \phi_s(x_s^i - w_s^i) + \sum_h \delta^{ih} k_h = 0 \end{aligned}$$

2. Each firm solves the following pb:

$$\max_{k^h} \quad -\phi_0 k^h + \sum_s \phi_s f^h(k_h; s)$$

3. Market clears[1]

$$\sum_i x^i \leq \sum_i w^i + \sum_h y^h$$

where $x^i \in \mathbb{R}_{+}^{L(S+1)}$.

4. Profits are

$$\pi^h = -\phi_0 k^h + \sum_s \phi_s f^h(k^h; s)$$

[1]Since $y^h = [-k^h, f^h(k^h)]$, this condition is the same as

$$\begin{aligned} \sum_i x_0^i &\leq \sum_i w_0^i - \sum_h k^h \\ \sum_i x_s^i &\leq \sum_i w_s^i + \sum_h f^h(k^h; s) \end{aligned}$$

6.3 Financial Market Equilibrium (with Production)

Full Model	**Simplified** ($J = 0, L = H = 1$)
Consumer solves $$\max_{x^i,\theta^i,z^i} u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i)$$ s.t. $p_0(x_0^i - w_0^i) + qz^i = (-p_0 k + Q)\theta_0^i - Q\theta^i$ $p_s(x_s^i - w_s^i) = \theta^i p_s f^h(k;s) + a_s z^i$	**Consumer solves** $$\max_{x^i,\theta^i,z^i} u^i(x_0^i) + \sum_s \text{Prob}_s u^i(x_s^i)$$ s.t. $(x_0^i - w_0^i) = (-k + Q)\theta_0^i - Q\theta^i$ $(x_s^i - w_s^i) = \theta^i p_s f^h(k;s)$
Firm solves $$\max_{k^h} \quad -p_0 k^h + Q(k^h)$$	**Firm solves** $$\max_{k^h} \quad -k^h + Q(k^h)$$
Market clears $$\sum_i x_0^i + \sum_h k^h \leq \sum_i w_0^i$$ $$\sum_i x_s^i \leq \sum_i w_s^i + \sum_h f^h(k^h;s)$$ $$\sum_i z^i = 0 \quad \text{and} \quad \sum_i \theta^i = 1$$	**Market clears** $$\sum_i x_0^i + \sum_h k^h \leq \sum_i w_0^i$$ $$\sum_i x_s^i \leq \sum_i w_s^i + \sum_h f^h(k^h;s)$$ $$\sum_i \theta^i = 1$$
Rational Conjectures $$Q^h(k^h) = \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$$	**Rational Conjectures** $$Q^h(k^h) = \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$$
Consistency in equity prices $$Q^h = Q^h(k^h), \quad \forall h \in H$$	**Consistency in equity prices** $$Q^h = Q^h(k^h), \quad \forall h \in H$$

Proposition 11. *Let (θ, k) be an equilibrium allocation. Consider an enlarged economy where agents can hold equity θ_e^i in the original firm $f(k)$ and γ_e^i in an off-equilibrium firm $f(k')$. Then:*

- *Under **Rational Conjectures**, $\gamma_e^i = 0$ for all i.*
- *Under **Drèze Conjectures**, the condition $\gamma_e^i = 0$ for all i is not guaranteed to hold (trade may occur).*

Proof (Sketch):

Mathematically, this result is because

$$\begin{aligned} & \mathbb{E}[m^i p \circ f^h(k^{h\prime})] \leq Q^h(k^{h\prime}), \quad \forall i \\ \exists j, \text{ such that} \quad & \mathbb{E}[m^j p \circ f^h(k^{h\prime})] > D^h(k^{h\prime}) \end{aligned}$$

In words, the proposition says:

- In the case where the equity is priced using the highest-bidder's evaluation, no one is going to buy the off-equilibrium firms if the highest-bidder doesn't buy it.
- However, if the equity of the off-equilibrium firms are priced using equilibrium holders' values, there might exist a potential buyer who values these firms more and thus want to buy them.

Proposition 12. *Assuming $J = 0, L = H = 1$, competitive equilibria of the FME with production are constrained P.E.*

The key of the proof relies on the optimization of the firm.

Proof (Sketch): Let (x, θ, k) be a competitive equilibrium. Suppose there exists $({x_0^i}', {\theta^i}') \succeq_P (x_0^i, \theta^i)$, then by the Law of Revealed Preference, we have

$$ {x_0^i}' + Q(k'){\theta^i}' \geq x_0^i + Q(k)\theta^i \forall i $$

where some i the inequality is strict. Now using the budget constraint and the consistency in the eqlm, $(x_0^i = w_0^i + (-k + Q(k))\theta_0^i - Q(k)\theta^i)$, we have

$$ \begin{aligned} {x_0^i}' + Q(k'){\theta^i}' &\geq x_0^i + Q(k)\theta^i \\ &= w_0^i + (-k + Q(k))\theta_0^i - Q(k)\theta^i + Q(k)\theta^i \\ &= w_0^i + (-k + Q(k))\theta_0^i \end{aligned} $$

Summing over i, and applying the market clearing condition ($\sum_i {\theta^i}' = 1$), then we have the following strict inequality

$$ Q(k') + \sum_i {x_0^i}' > -k + Q(k) + \sum_i w_0^i $$

Now here comes the critical step: Due to the optimization of the firm, we have

$$ -k + Q(k) \geq -k' + Q(k') $$

It follows that

$$ Q(k') + \sum_i {x_0^i}' > -k' + Q(k') + \sum_i w_0^i $$

Hence, we derive a contradiction: $\sum_i {x_0^i}' + k' > \sum_i w_0^i$. ∎

6.4 Rational price conjectures

Proposition 13. *Define* $Q^h(k^h) := \max_i \mathbb{E}[m^i p \circ f^h(k^h)]$*, and* $D^h(k^h) := \mathbb{E}[\sum_i \theta^i_h m^i p \circ f^h(k^h)]$*, we have*

$$\text{FME with Rational Conjectures } (Q^h) \quad \overset{\Rightarrow}{\not\Leftarrow} \quad \text{FME with Dreze Conjectures} (D^h)$$

Proof (Sketch): Note that in equilibrium, these equity prices coincide with RC and Dreze Conjecture, namely, $Q^h(k^{h^*}) = D^h(k^{h^*})$, where k^{h^*} is the equilibrium input choice of firm h. However, away from equilibrium, we have

$$Q^h(k^h) \geq D^h(k^h)$$

Hence, there are no profitable deviations from equilibrium under RC, (that is the $\Rightarrow$ part), but there could be profitable deviations under Dreze (that is the $\not\Leftarrow$ part). ■

6.5 Corporate Finance

6.5.1 FME with bonds and equity

Definition. A competitive equilibrium of a Financial Market Economy with production, rational price conjecture, under 2 periods (assuming $I > 1$ individuals, S states, $J = 1$ assets, and $H = 1$ firm) is defined by prices and quantities $(x, z, \theta, B, k, q, Q, P) \in \mathbb{R}^{L(S+1)I}_+ \times \mathbb{R}^{JI}_+ \times \mathbb{R}^{HI}_+ \times \mathbb{R}^{HI}_+ \times \mathbb{R}^{LH}_+ \times \mathbb{R}^{J}_{++} \times \mathbb{R}^{H}_{++} \times \mathbb{R}^{H}_{++}$, such that

1. Each consumer solves

$$\begin{aligned} \max_{x^i, z^i, \theta^i, B^i} \quad & u^i(x^i_0) + \sum_{s \in S} \text{prob}_s u^i(x^i_s) \\ \text{subject to:} \quad & x^i_0 - \omega^i_0 = (-k + Q + PB)\,\theta^i_0 - Q\theta^i - PB\theta^i - qz^i \\ & x^i_s - \omega^i_s = a_s z^i + \theta^i(f(k^h; s) + B), \forall s \in S \end{aligned}$$

2. Each firm solves

$$\max_{k^h, B^h} \quad - k^h + Q(k^h, B^h) + P(k^h, B^h)B^h$$

3. Markets clear

$$\text{Goods:} \quad \sum_i (x^i - w^i) = 0$$
$$\text{Assets:} \quad \sum_i z^i = 0$$
$$\text{Equity:} \quad \sum_i \theta^i = 1$$

4. The price of equity satisfies rational conjectures (for in- and off- equilibrium paths)

$$Q^h(k^h, B^h) = \max_i \mathbb{E}\Big[m^i[\max\{0, f(k;s) - B\}]\Big], \quad \forall h$$
$$P^h(k^h, B^h) = \max_i \mathbb{E}\Big[m^i[\min\{1, \frac{f(k;s)}{B}\}]\Big], \quad \forall h$$

5. Consistency: $Q^h = Q^h(k^h, B^h)$ and $P^h = P^h(k^h, B^h)$.

Chapter 7

Assignment and Search

Proposition 14. *The dual assignment and the primal assignment problems yield the same solution.*

Proof The Primal problem is given by

$$\begin{aligned} \max_{\pi^{ih}} \quad & \sum_i \sum_h \pi^{ih} U^{ih} \\ \text{s.t.} \quad & \sum_h \pi^{ih} = w^i, \quad \forall i \\ & \sum_i \pi^{ih} = f^h, \quad \forall h \end{aligned}$$

The Dual problem is given by

$$\begin{aligned} \min_{w^i, f^h} \quad & \sum_i u^i w^i + \sum_h v^h f^h \\ \text{s.t.} \quad & u^i + v^h \geq U^{ih}, \quad \forall i, h \end{aligned}$$

The Lagragean for the Primal problem is

$$\mathcal{L}^P = \sum_i \sum_h \pi^{ih} U^{ih} + \sum_h v^h \left(f^h - \sum_i \pi^{ih} \right) + \sum_i u^i \left(w^i - \sum_h \pi^{ih} \right)$$

The Lagragean for the Dual problem is

$$\mathcal{L}^D = \sum_i u^i w^i + \sum_h v^h f^h + \sum_i \sum_h \pi^{ih} \left(U^{ih} - u^i - v^h \right)$$

Obivously, when choosing proper Lagragean Multipliers, namely, using $\{v^h\}, \{u^i\}$ as the

LM for the Primal problem and $\{\pi^{ih}\}$ for the Dual problem, these two Lagragean coincide with each other and thus yield the same solutions. ■

Proposition 15. *If U^{ih} is supermodular, then the optimal matching is assortative.*

Proof Mathematically, it is a direct implication of the Hardy-Littlewood-Polya inequality, which says

$$\langle x, y\rangle \leq \langle \hat{x}, \hat{y}\rangle,$$

where $\hat{x}, \hat{y}$ are the decreasing rearrangements of x, y respectively.

Now, we prove the proposition by contradiction with inductive steps. Assume that the optimal mathching is not assortative; concretely, assume that worker i matches with firm $\sigma_*(i) \neq i$. Then, using the supermodularity of U, we have

$$\sum_{i\neq 1\neq {\sigma_*}^{-1}(1)} U^{i\sigma_*(i)} + U^{11} + U^{{\sigma_*}^{-1}(1)\sigma_*(1)} > \sum_i U^{i\sigma_*(i)}.$$

In the k^{th} step, we have

$$\sum_{i\neq k\neq {\sigma_*}^{-1}(k)} U^{i\sigma_*(i)} + U^{kk} + U^{{\sigma_*}^{-1}(k)\sigma_*(k)} > \sum_i U^{i\sigma_*(i)}.$$

Iterate this procedure for all i, we reach a contradiction. ■

Chapter 8

Asymmetric Information

8.1 Prescott-Townsend Competitive Equilibrium

Definition. A competitive equilibrium in a Moral Hazard Economy is defined by an allocation $(x, e) \in \mathbb{R}^S \times E$, and prices $q \in \mathbb{R}^{SE}$ such that

1. Each consumer i solves

$$\begin{aligned} \max_{x^i, e} \quad & \sum_s \text{Prob}_s(e) u(x_s) - v(e) \\ \text{subject to:} \quad & \sum_s q_s(e)(x_s - w_s) = 0 \end{aligned}$$

2. Market clears

$$\sum_s \text{Prob}_s(e)(x_s^i - w_s^i) = 0,$$

   where due to the Law of Large Number, $\text{Prob}_s(e)$ is the fraction of consumers choose effort e when state s is realized.

3. Self-imposing incentive constraint:

$$e \in \text{argmax}_\epsilon \sum_s \text{Prob}_s(\epsilon) u(x_s) - v(\epsilon), \quad \text{given} x$$

4. Rational conjecture

$$\text{Prob}_s(e) = q_s(e)$$

8.2 Radner Rational Expectation Equilibrium

For an economy with $t = 0, 1$, S states, J assets, $L = 1$ commodity, $I > 1$ agents with strictly positive endowments and strictly monotonic, vNM, and strictly quasiconcave preferences, and Σ^I signals. A Radner Rational Expectation Equilibrium consists of $(x, z, q_\sigma, \phi) \in \mathbb{R}_+^{(S+1)I} \times \mathbb{R}_+^{JI} \times \mathbb{R}_{++}^{J|\Sigma|^I} \times \Phi$ such that

1. Given $w_i, A, q_\sigma, \phi, \sigma_i$, agent i solves the following problem

$$\begin{aligned} \max_{x_i, q_i} \; & U_i(x_0^i) + \sum_s \text{prob}_s^{\sigma_i, \phi^{-1}(q_\sigma)} U_i(x_s^i) \\ \text{subject to: } & x_0^i - w_0^i + q_\sigma z_i = 0 \\ & x_s^i - w_s^i = a_s z_i, \quad \forall s \end{aligned}$$

   where $\text{prob}_s^{\sigma_i, \phi^{-1}(q_\sigma)}$ is the posterior beliefs about the state s given i's signal σ_i and her information about others' signals given the inverse price-map and the price $\phi^{-1}(q_\sigma)$.

2. Markets clear

   - Assets (signal-dependent): $\sum_i z_i(q_\sigma, \phi) = 0, \forall \sigma \in \Sigma^I$
   - Goods at $t = 0$: $\sum_i x_0^i - w_0^i = 0$
   - Goods at $t = 1$: $\sum_i x_s^i - w_s^i = 0, \forall s$

3. Assets prices are consistent: $q_\sigma = \phi(\sigma)$.

8.2.1 Fully Revealing Equilibria

Definition 8.2.1. A REE price map $\phi : \Sigma^I \to \mathbb{R}_+^J$ is *fully revealing* if

$$\phi(\sigma) \neq \phi(\sigma'), \quad \text{for any } \sigma, \sigma' \in \Sigma^I \text{ with } \sigma \neq \sigma'.$$

Equivalently, ϕ is invertible: by observing q_σ, every agent can back out the joint signal profile $\sigma \in \Sigma^I$, so the posterior conditional on $(\sigma^i, \phi^{-1}(q_\sigma))$ collapses to the full-information posterior conditional on σ,

$$\text{prob}_s^{\sigma^i, \phi^{-1}(q_\sigma)} = \text{prob}_s^\sigma.$$

Theorem 2 (Radner 1979)**.** *For a generic subset of economies parametrized by endowments $\omega \in \mathbb{R}_{++}^{(S+1)I}$, there exists a REE price map $\phi : \Sigma^I \to \mathbb{R}_+^J$ that is fully revealing.*

Proof sketch (counting equations and unknowns). Because a fully revealing ϕ is invertible, every agent's posterior is the full-information posterior prob_s^σ. Consider the auxiliary economy in which agent i's problem is

$$\begin{aligned}
\max_{(x^i,z^i)\in\mathbb{R}_+^{S+1}\times\mathbb{R}^J} \quad & u^i(x_0^i) + \sum_{s\in S} \text{prob}_s^\sigma\, u^i(x_s^i) \\
\text{subject to:} \quad & x_0^i + q_\sigma z^i \leq \omega_0^i, \\
& x_s^i \leq a_s z^i + \omega_s^i, \quad \forall s.
\end{aligned}$$

Let $z^i(q_\sigma, \phi)$ denote the optimal portfolio and $Z(q_\sigma, \phi) = \sum_{i\in I} z^i(q_\sigma, \phi)$ the aggregate excess demand for the J assets.

We argue by contrapositive: suppose ϕ is *not* fully revealing, i.e., there exist $\sigma \neq \sigma' \in \Sigma^I$ such that $\phi(\sigma) = \phi(\sigma')$. Then the following three conditions must hold simultaneously:

$$\begin{aligned}
Z(\phi(\sigma), \phi) &= 0, \\
Z(\phi(\sigma'), \phi) &= 0, \\
\phi(\sigma) &= \phi(\sigma').
\end{aligned}$$

Count: treating $\phi(\sigma), \phi(\sigma') \in \mathbb{R}^J$ as the unknowns, this is a system of $3J$ equations in $2J$ unknowns. By the Transversality Theorem applied to the endowment parameter $\omega \in \mathbb{R}_{++}^{(S+1)I}$, an over-determined system of this form generically (off a closed, measure-zero subset of ω) has no solution. Hence generically no two distinct signal profiles deliver the same equilibrium price, and the REE is fully revealing. $\square$

Remarks.

- At a fully revealing REE, all agents share the same posterior prob_s^σ, so the equilibrium allocation is exactly the one in a full-information financial market economy: *agents do not trade on account of information*, only on account of risk-sharing.

- The genericity logic is the same as the one used in incomplete markets (see Geanakoplos-Polemarchakis): a transversality count $3J$ vs. $2J$ pins down the result, even though we have not made the perturbation argument explicit.

- The result *fails* when the signal space is too rich relative to the asset span. If $\sigma \in \mathbb{R}^{\Sigma I}$ is continuous and $\Sigma I > J$, the price map $\phi : \mathbb{R}^{\Sigma I} \to \mathbb{R}^J$ generically cannot be invertible by a dimensionality argument: ϕ maps a ΣI-dimensional set into a J-dimensional one (Theorem 6.2 / Allen 1981).

8.3 Moral Hazard in Production

Chapter 9

Questions:

- What are examples in the literature that show that REE prices might not exist for some economies (the first example is due to D. Kreps)?
- How do we show a production economy with unobserved choices of risk ϕ, $f^h(k^h, \phi)$, presents Moral Hazard?
- Are the following true?
  - With bid-ask spread (i.e., transaction costs), $q \leq \pi A \leq q + \gamma$? where $q = \pi A$ if $\gamma = 0$.
- Why do we have the following statements?
  - (pp. 99) "allowing negative position in firm equity makes small firms change the asset span"
  - In equilibirum, denote the equilibrium input as k^{h^*}, then $Q^h(k^{h^*}) = D^h(k^{h^*})$. In words, prices are the same in equilibrium for both RC and Dreze.
- Confusions about the proofs:
  - I don't see how Lemma 5.1 is used in the proof of Theorem 5.1.
- How to define a competitive equilibrium with bonds and equity? In particular, how to specify the Bond at time 0?